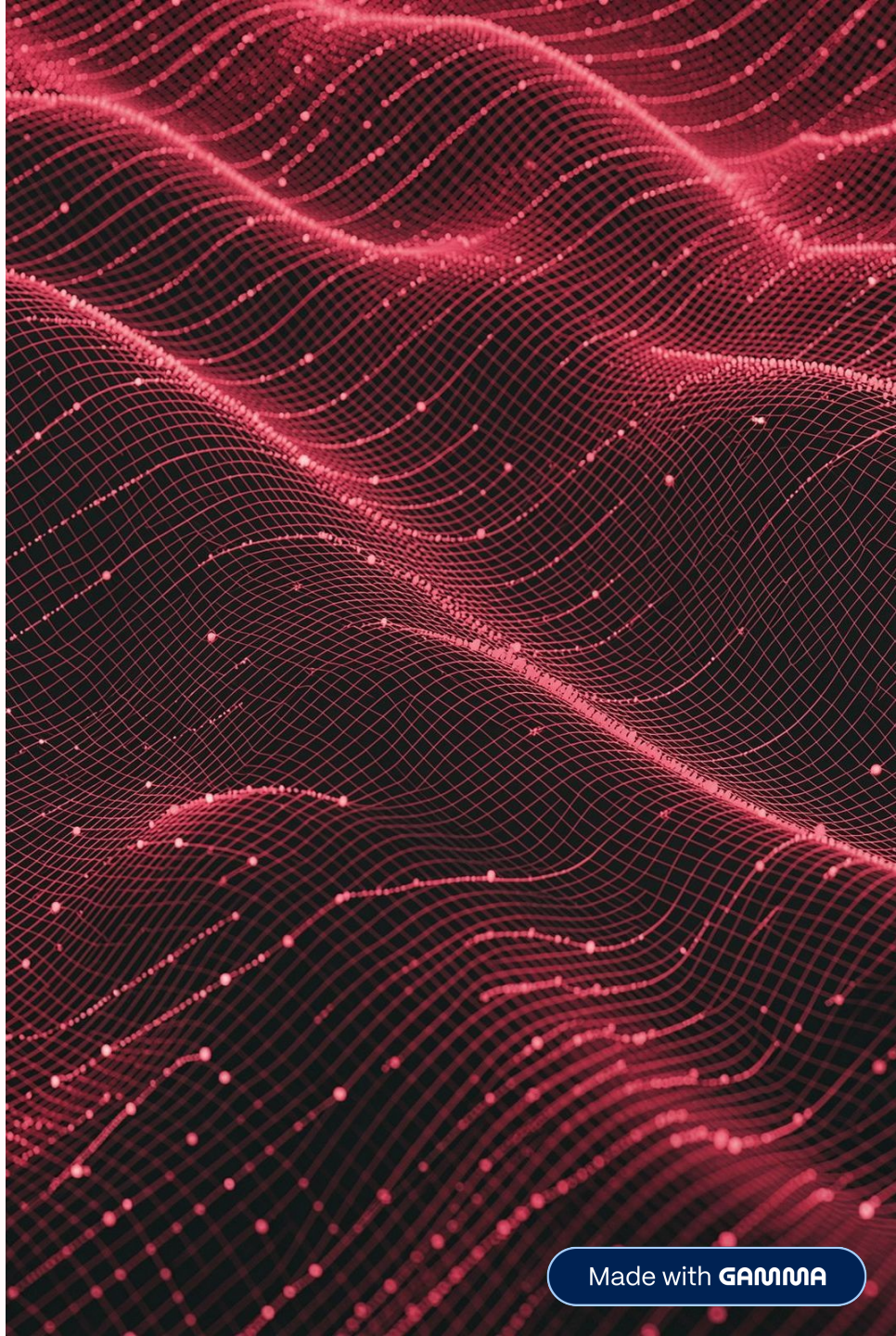
Object Detection of Unaware Pedestrians in Warehouse Environments

Helpforce.AI Research Program · Isaac Sim Robotics Simulation · Perception Team — Phase 1



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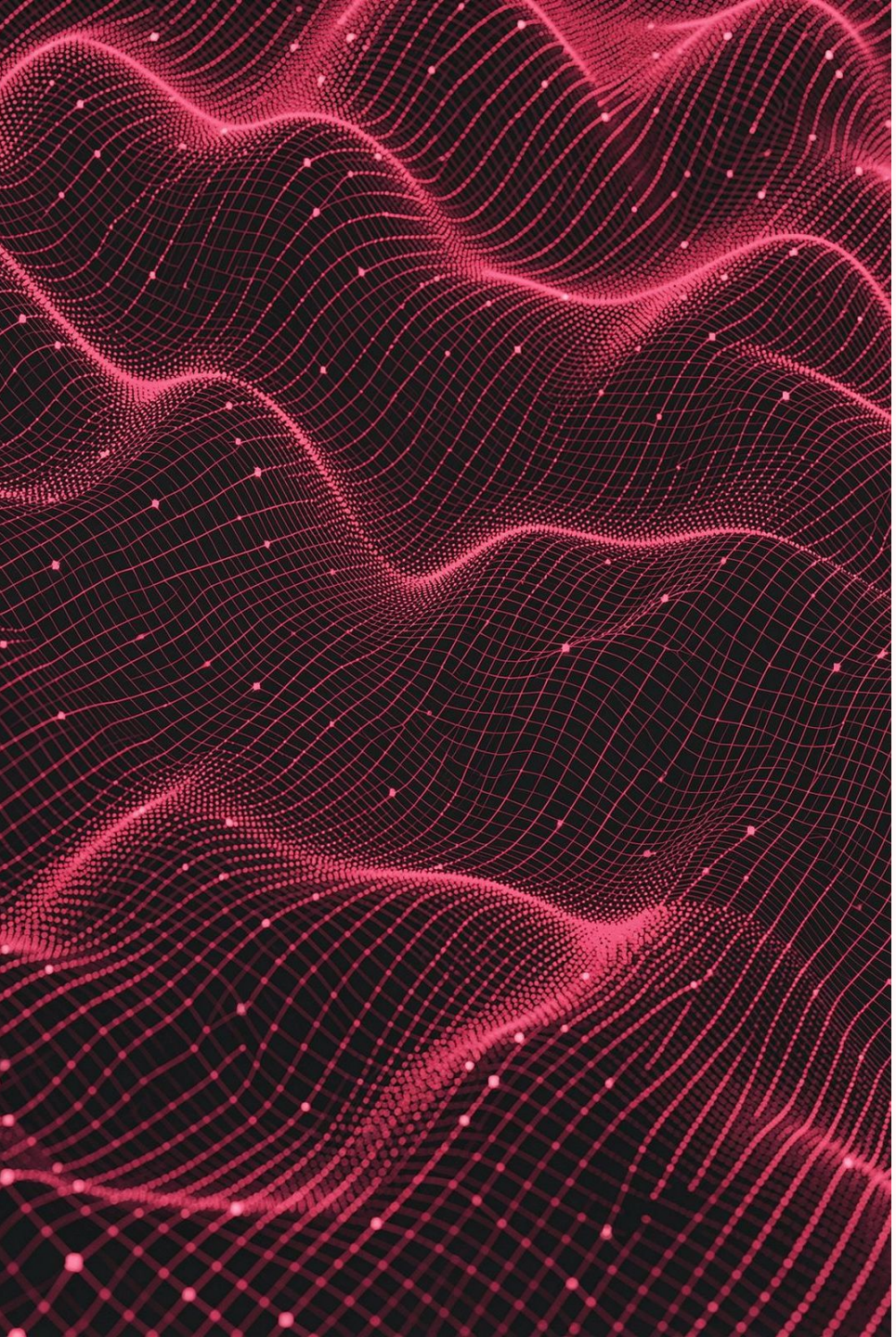

Project Scope & Objectives

Goal

Develop **robust pedestrian detection** for mobile robots navigating warehouse environments with partial occlusions — prioritizing safety for unaware personnel.

Focus Areas

Sensor selection rationale, data pipeline design, and algorithm integration within the **Isaac Sim** simulation environment.

Phase 1 Constraint

Pre-built mobile robot assets only — **no custom hardware implementation**. Simulation-first approach using Isaac Sim's built-in asset library.

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Sensor Modality Comparison

Trade-off analysis across four candidate sensor types for warehouse pedestrian detection:

Modality	Strengths	Weaknesses	Warehouse Relevance	Verdict
RGB Camera	High resolution, color/texture, large DL training sets	Struggles in low light, computationally intensive	Essential for visual pedestrian features — pose, gait, clothing	✅ Primary
LiDAR	3D point clouds, lighting-robust, precise ranging	Limited texture, high cost, lower angular resolution	Occlusion detection, 3D bounding boxes, reliable in dark aisles	✅ Secondary
Radar	Weather-robust, measures velocity directly	Low spatial resolution, poor fine-grained discrimination	Useful for velocity; insufficient for pedestrian classification	⚠️ Limited
Ultrasonic	Low cost, no interference	~5m range cap, very low spatial resolution	Proximity alerts only; inadequate for detection pipeline	❌ Excluded

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Sensor Selection Rationale

Primary: RGB Camera

Essential for visual pedestrian cues — pose, clothing, and gait. Enables deployment of deep learning models trained on large-scale human datasets (COCO, WiderPerson). Highest information density per frame for classification tasks.

Secondary: LiDAR

Detects occluded pedestrians through point cloud geometry analysis. Provides **3D bounding boxes independent of lighting conditions** — critical in warehouse environments with uneven illumination and shadowed aisles.

Fusion Strategy

Camera and LiDAR operate as complementary modalities with distinct responsibilities:

- **Camera** → Classification, feature extraction, visual identity
- **LiDAR** → Localization, depth estimation, occlusion handling
- **Fusion layer** → Late fusion at bounding box level or early fusion at feature map level

This division of responsibility ensures the system degrades gracefully — if one sensor fails, the other provides fallback detection.

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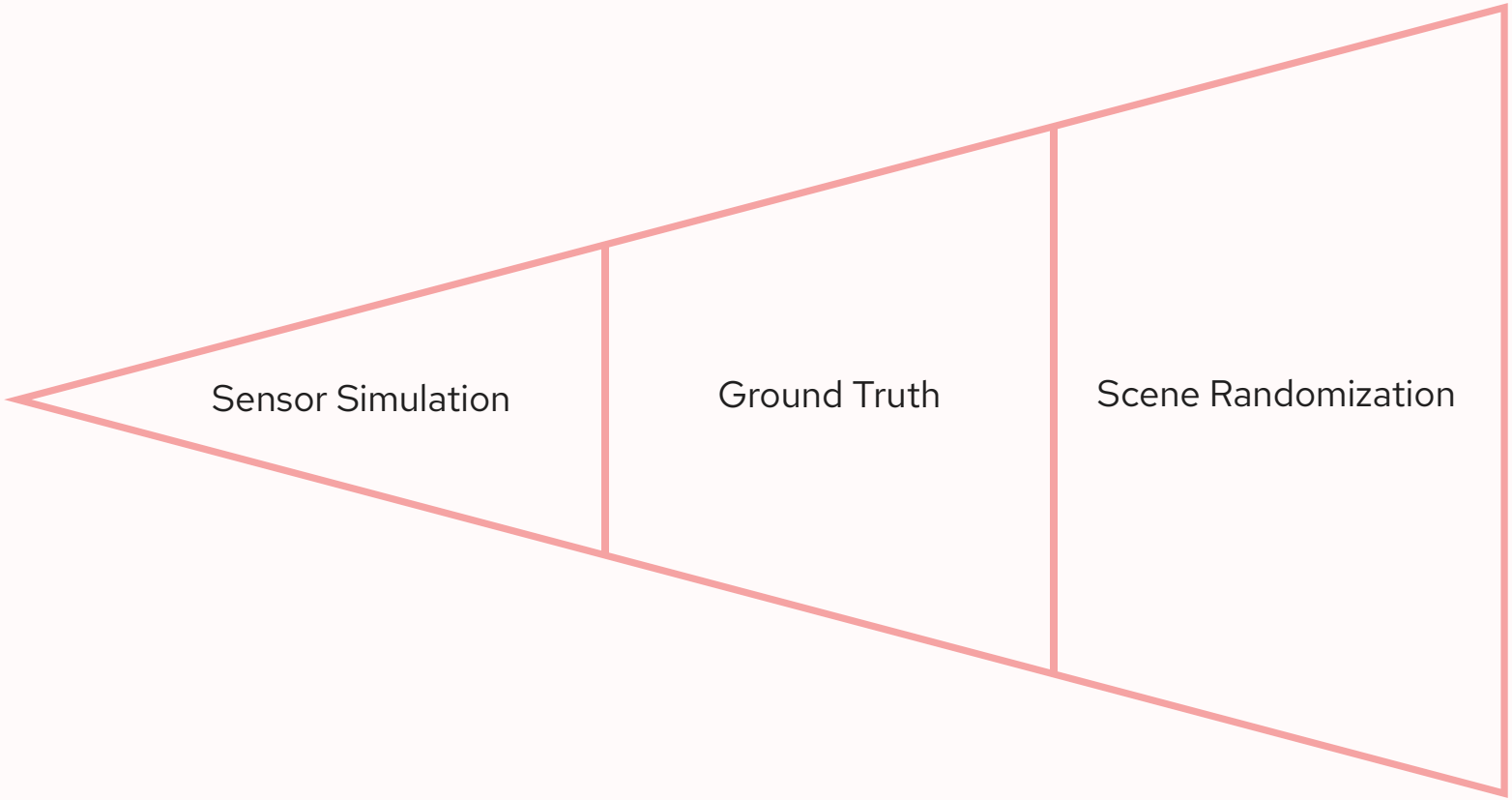
Data Acquisition Pipeline in Isaac Sim



Synchronized multi-sensor capture ensures temporal alignment between camera frames and LiDAR scans. Ground truth is auto-generated — eliminating manual annotation overhead for synthetic training data.

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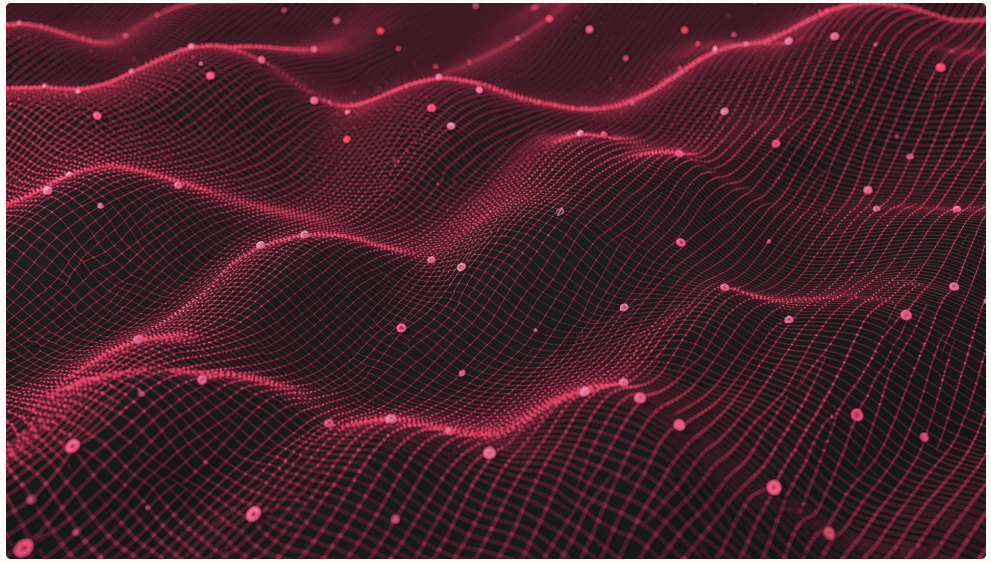

Image Preprocessing & Feature Extraction

Camera Stream

- Resize to model input resolution (e.g., `416×416` for YOLO)
- Normalize pixel values to [0, 1] range
- Augmentation: rotation, brightness jitter, horizontal flip

LiDAR Processing

- Voxelize point clouds to reduce computational load
- Ground plane removal via RANSAC or height thresholding
- Extract height and density features per voxel column

Temporal Fusion

Stack consecutive frames to capture **motion cues** — distinguishes mobile pedestrians from static warehouse objects at the feature level.

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Detection & Segmentation Techniques

Bounding Box Detection

Localizes pedestrian position and size in the image frame. Enables **distance estimation** for collision avoidance buffers and robot deceleration triggers.

Instance Segmentation

Pixel-level masks per pedestrian. **Critical for partial occlusions** — correctly attributes overlapping pedestrians in crowded aisle scenarios.

Pose Estimation

Keypoint detection across body joints infers **pedestrian orientation and movement intent** — enables predictive path planning for the robot.

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Algorithm Comparison: YOLO vs Cosmos vs Isaac Built-ins

YOLO v8/v10

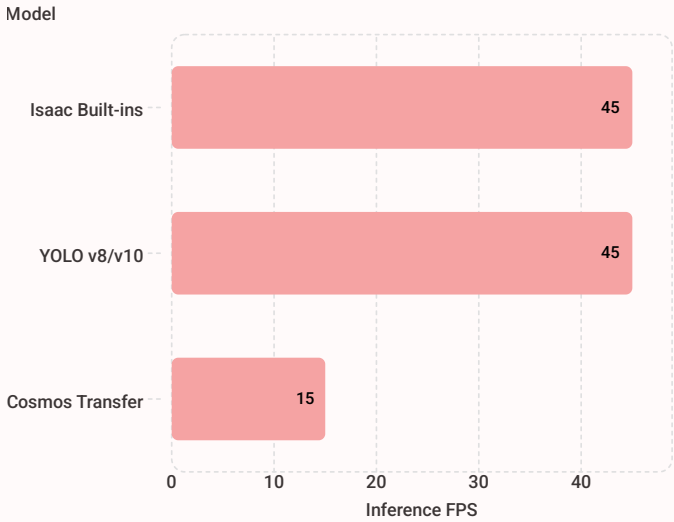
30–60 FPS real-time throughput. Requires custom warehouse training data (~5k annotated images). Excellent on unoccluded targets; performance degrades with heavy occlusion. Best for **speed-critical deployments**.

Cosmos Transfer Learning

~15 FPS inference. Foundation model pre-trained on diverse pedestrian datasets. Faster domain adaptation with fewer labeled examples. **Superior occlusion generalization**. Higher GPU memory footprint.

Isaac Sim Built-ins

Pre-trained models optimized for simulation environments. **Zero training time** — instant deployment. Limited customization; may underperform on real-world warehouse conditions outside synthetic distribution.



FPS estimates are approximate; actual throughput depends on GPU hardware and input resolution.

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YOLO Architecture Deep Dive

1

Backbone

CSPDarknet extracts **multi-scale feature maps** from the input image — captures low-level edges and high-level semantic patterns simultaneously.

2

Neck (FPN/PAN)

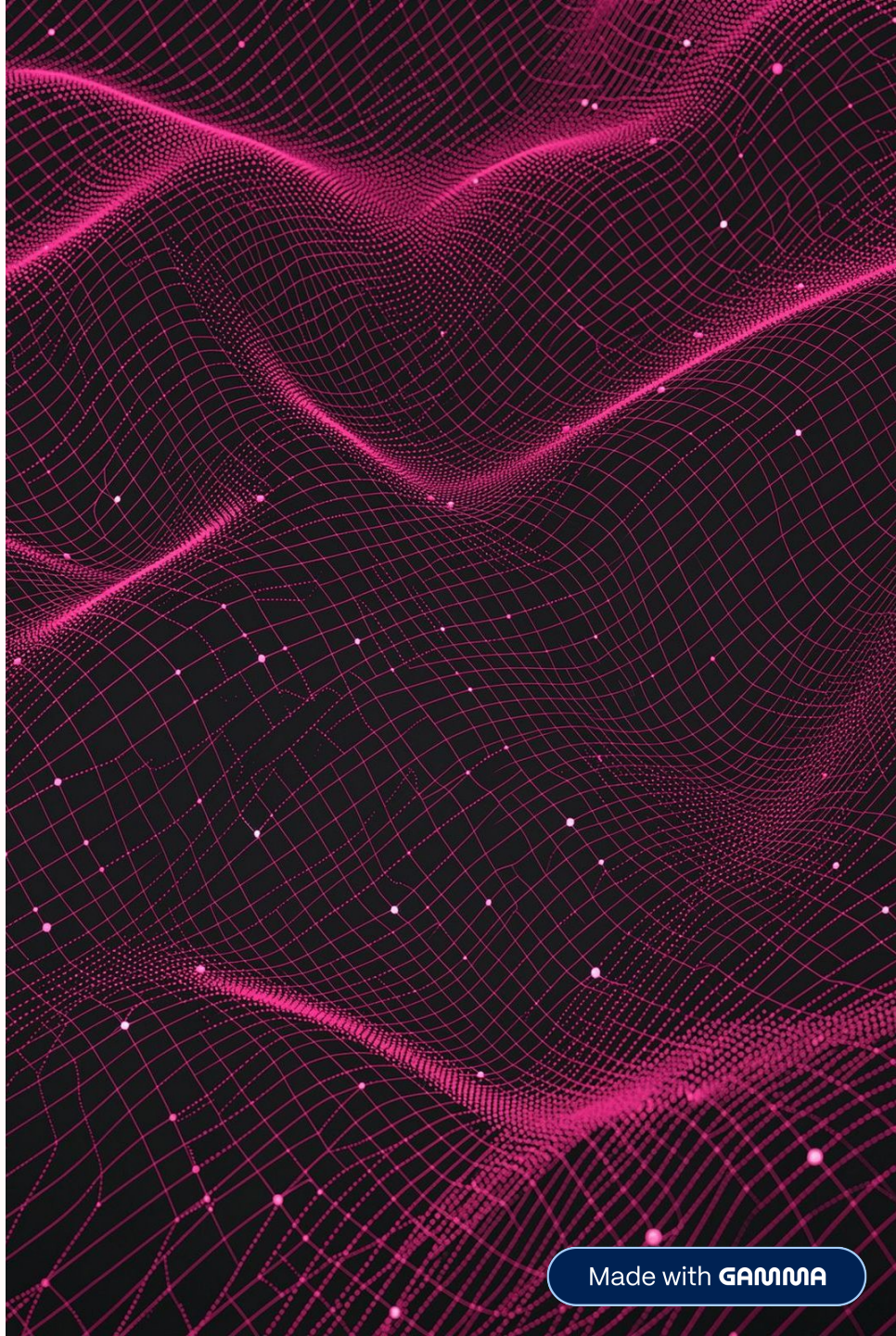
Feature Pyramid Network fuses representations across scales — enables detection of **pedestrians at varying distances** and apparent sizes within the same frame.

3

Detection Head

Predicts class probabilities, bounding box coordinates, and confidence scores per anchor. Outputs decoded as **NMS-filtered bounding boxes** per pedestrian instance.

Training recommendation: Fine-tune on ~5k annotated Isaac Sim synthetic images before any real-world transfer to minimize domain gap.



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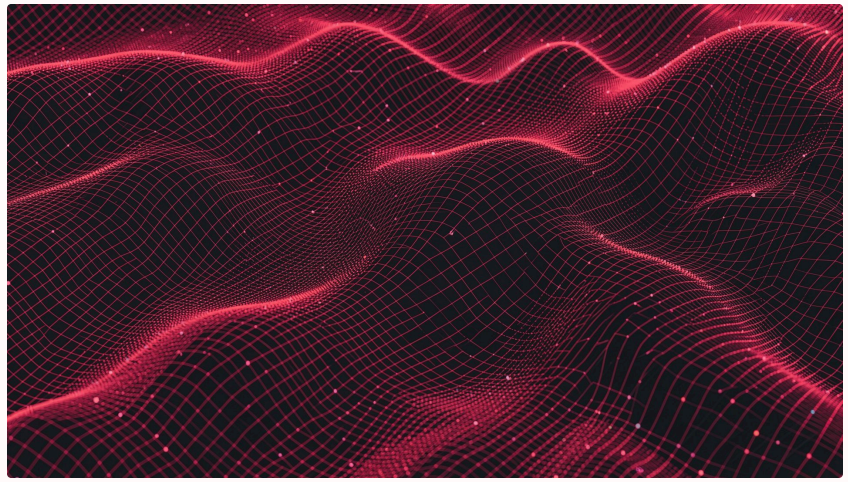
Cosmos Transfer Learning Approach



Foundation Model

Pre-trained on diverse pedestrian datasets encoding **rich semantic understanding** of human shape, clothing variation, and body proportion — reducing the cold-start problem for warehouse-specific detection.

Domain Adaptation

Fine-tune on warehouse-specific synthetic data from Isaac Sim with **minimal labeled examples**. Cosmos's pre-trained representations compress the annotation burden significantly compared to training from scratch.

Key Trade-offs

- **Advantage:** Handles partial and full occlusions through learned context priors
- **Advantage:** Reduces annotation burden; fewer labeled examples required
- **Trade-off:** ~15 FPS inference — slower than YOLO for real-time applications
- **Trade-off:** Requires larger GPU memory allocation for foundation model weights

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Isaac Sim Perception Stack

Built-in Detectors

Pre-trained person detection models deployable **without any training step**. Ideal for rapid prototyping and baseline benchmarking within Phase 1 timeline constraints.

Synthetic Data Generation

Domain randomization across **lighting, textures, pedestrian poses, and occlusion patterns**. Programmatic scene variation creates training diversity without real-world data collection.

ROS Integration

Detection results published as `geometry_msgs/PoseArray` for downstream consumption by control and planning modules — plug-and-play with standard ROS navigation stacks.

Known Limitation

Limited customization depth. **Sim-to-real gap** may degrade performance outside the synthetic distribution — real-world validation essential before production deployment.

⚠ Next steps to confirm: GPU inference latency targets, real-world validation protocol, and Phase 2 hardware integration requirements.

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